Integration of reciprocal functions

1 $f(x) = \ln x + C, x > 0$

2

a $-2\ln x + C, x > 0$

b $6\ln x + C, x > 0$

c $\frac{1}{7}\ln x + C, x > 0$

d $\ln(x+5) + C, x > -5$

3

a $4\ln x + C, x > 0$

b $-7\ln x + C, x > 0$

c $\frac{3}{4}\ln x + C, x > 0$

d $\ln(x-4) + C, x > 4$

e $2\ln(x+3) + C, x > -3$

f $\ln(2x+3) + C, x > \frac{-3}{2}$

g $-2\ln(x-4) + C, x > 4$

h $\frac{1}{2}\ln(2x+3) + C, x > -\frac{3}{2}$

i $-\ln(5-x) + C, x < 5$

j $\frac{2}{3}\ln(3x+5) + C, x > -\frac{5}{3}$

k $\ln(x^2-5) + C, x < -\sqrt{5}, x > \sqrt{5}$

l $8\ln(x) + 5x + C, x > 0$

4

a $3\ln 3$

b $\ln 2$

c $2\ln\frac{7}{5}$

d $\frac{4}{3}\ln\frac{5}{2}$

e $5\ln 2$

f $8 + \ln\frac{5}{3}$

g $\frac{1}{4}\ln\frac{17}{5}$

h $32 + \ln 2$

5

a $x^2 + x - 3\ln x + C, x > 0$

b $5x + 2\ln x + C, x > 0$

6

a $f'(x) = 1 + \ln x$

b $x\ln x - x + C, x > 0$

Reverse chain rule

7

a $f'(x) = \frac{2x}{x^2-4}$

b $3\ln(x^2-4) + C$

8

a $f'(x) = \frac{2x+2}{x^2+2x}$

b $\frac{1}{4}\ln(2x^2+4x) + C$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

9

a $f'(x) = \dfrac{9x^2 + 10x}{3x^3 + 5x^2}$

b $2\ln\left(3x^3 + 5x^2\right) + C$

10

a $\ln(4x - 3) + C$

b $5\ln(3x - 2) + C$

c $-4\ln(2x + 1) + C$

d $2\ln\left(x^2 + 1\right) + C$

e $\ln\left(x^3 + 1\right) + C$

f $2\ln\left(2x^2 - 3\right) + C$

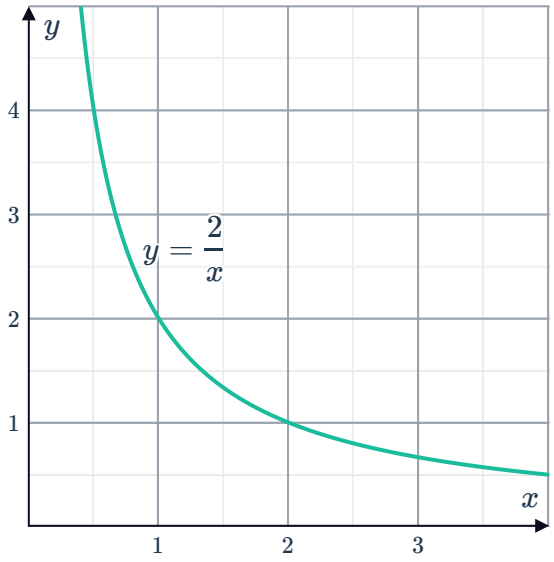
g $\dfrac{1}{3}\ln\left(2x^3 + 7\right) + C$

h $\ln\left(2x^3 + 3x\right) + C$

Area under the curve

11

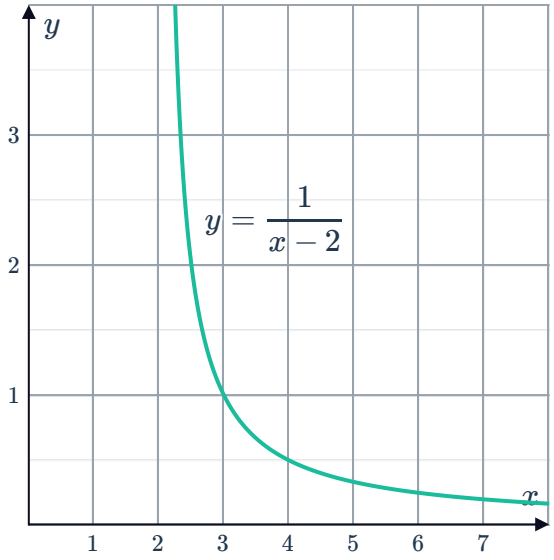
a



b $2\ln\dfrac{3}{2}$ units^2

12

a

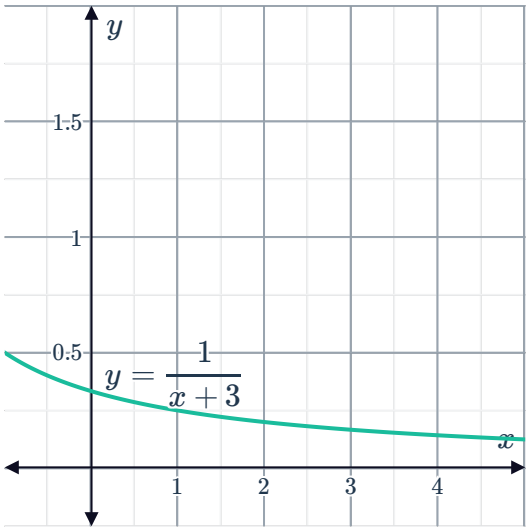


b $\ln 3$ units^2

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

13

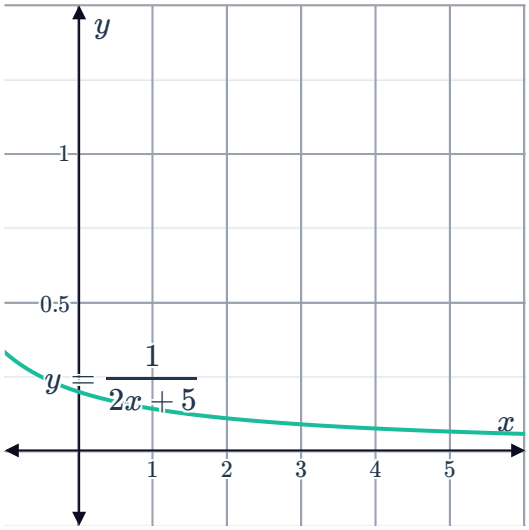
a



b $\ln \frac{7}{5}$ units^2

14

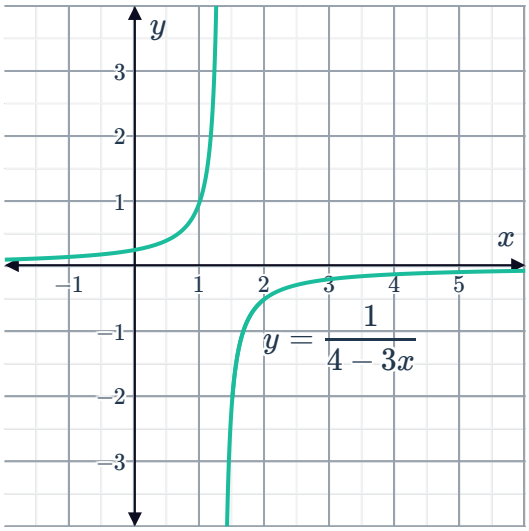
a



b $\frac{1}{2} \ln \frac{13}{11}$ units^2

15

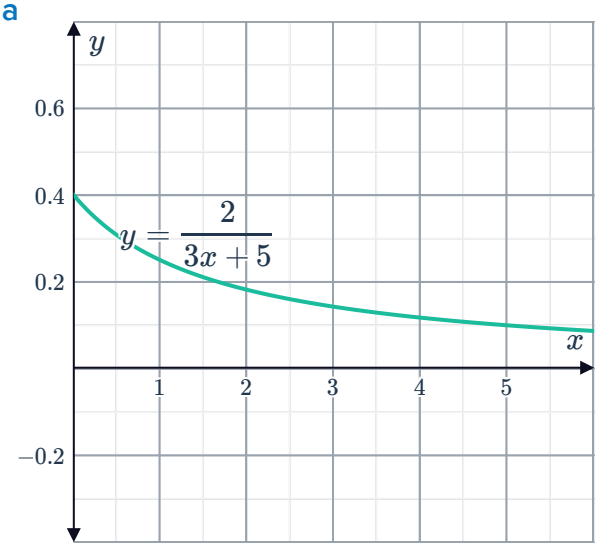
a



b $\frac{1}{3} \ln \frac{11}{5}$ units^2

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

16 **a**



b $\frac{2}{3}\ln\frac{17}{5}$ units^2

17

a $(-\infty, 0) \cup (0, \infty)$

b $y \to \infty$

c $(70 + \ln 6)$ units^2

18 $2 + 5\ln\frac{7}{5}$ units^2

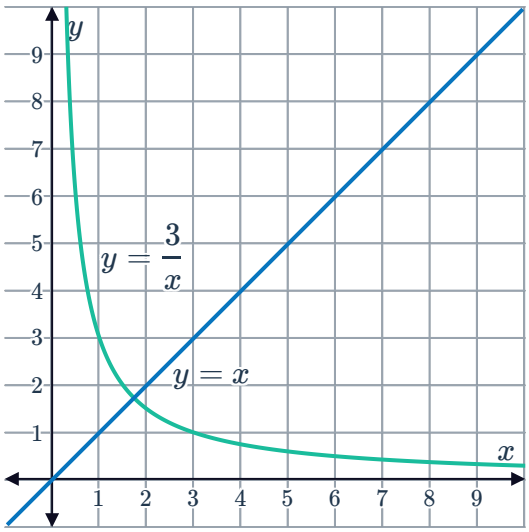
19

a $\frac{3x^4}{4} - 2x + \frac{1}{2}\ln x + C$

b $\frac{37}{4} + \frac{\ln 2}{2}$ units^2

20

a



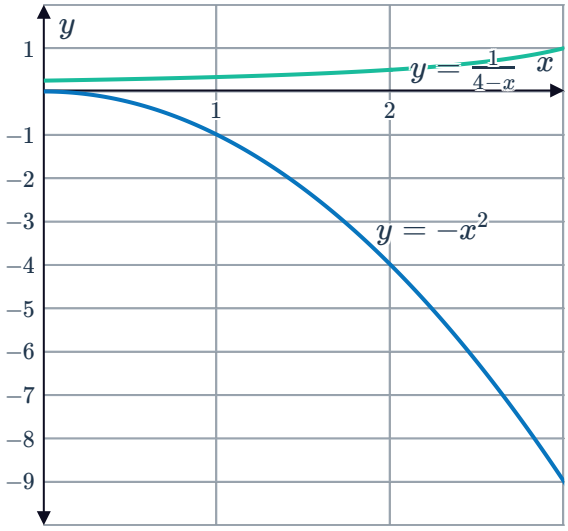
b $x = \sqrt{3}$

c $\frac{3}{2} + 3\ln\sqrt{3}$ units^2

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

21

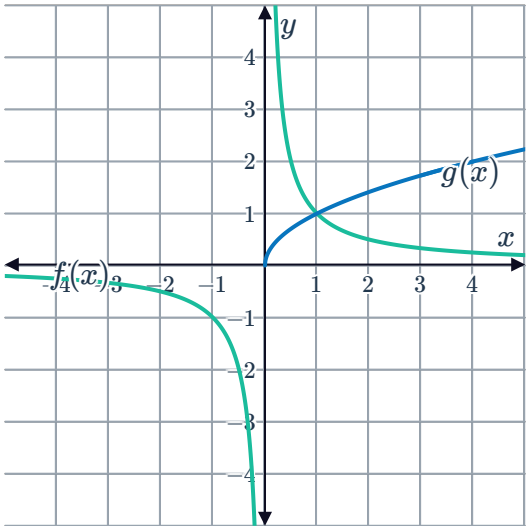
a



b $\left(\frac{26}{3} + \ln 3\right)$ units^2

22

a



b $x = 1$

c $\ln 2 + \frac{1}{3}$ units^2

23 $k = e^{\frac{1}{3}}$

24

a $f'(x) = \frac{2x}{x^2+1}$

b $\frac{1}{2}\ln\left(x^2+1\right) + C$

c $\frac{1}{2}\ln 2$ units^2

25

a $f'(x) = \frac{\cos x}{\sin x + 1}$

b $\ln(\sin x + 1) + C$

c $\ln\left(\frac{3}{2}\right)$ units^2

© ClipSAT: By Mr. Mohamed Abdallah +966597688647